COMPUTER HARDWARE ASSIGNMENT

Desktop PC: Components, Disassembly & Reassembly

Components | Step-by-step diagrams | Safety checklist

Contents

1	**Main components inside a desktop PC** Every part you see when the cabinet is opened, with a short description.
2	**Disassembly steps with labelled diagrams** 10 steps from unplugging the power to removing the motherboard.
3	**Reassembly steps and why the order matters** 9 steps to build the PC back up safely and correctly.
4	**Safety precautions and tools checklist** What to prepare before you start, and why each item matters.

Big picture: A desktop PC is built from a few main parts that each do one job: the **CPU** thinks, the **RAM** holds data for quick use, the **storage** keeps files, the **GPU** draws the picture, the **PSU** gives power, and the **motherboard** connects everything. Keep this picture in mind while reading the steps.

About the pictures: I cannot take real photos for you, so every step uses a labelled diagram. Highlighted (gold) parts show what is being worked on. If your teacher wants real photos, take one at each step and paste it next to the matching diagram, using the same captions.

QUESTION 1 Main components inside a desktop PC

After you remove the side panel of the cabinet (also called the case), you will see the following parts. The diagram shows where they usually sit.

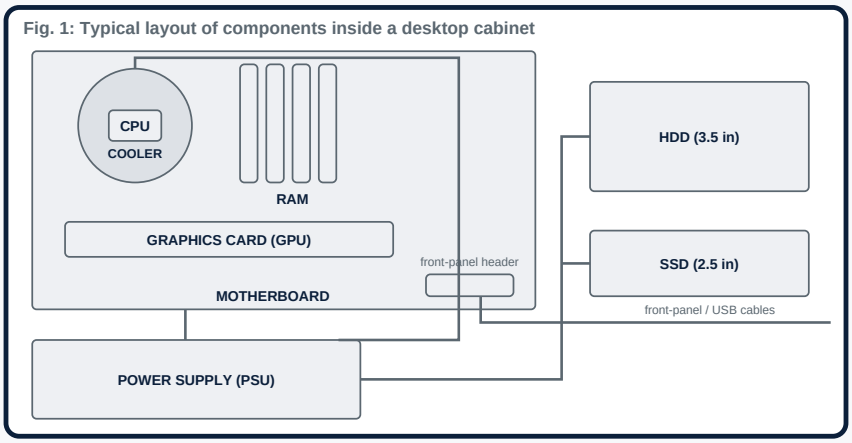

Fig. 1: Typical layout inside a desktop cabinet (exact positions differ between models).

No.	Component	Short description
1	**Motherboard**	The large main circuit board. Every other part plugs into it, and it lets them talk to each other.
2	**CPU (Processor)**	The 'brain' of the PC. It runs program instructions and does the calculations. It sits in a square socket on the motherboard.
3	**CPU Cooler (Heatsink + Fan)**	A metal block with a fan placed on the CPU. It removes heat so the processor does not overheat.
4	**RAM (Memory)**	Thin sticks in long slots next to the CPU. They store data temporarily for fast access while the PC is on. Data is lost when power is off.
5	**Graphics Card (GPU)**	A card in the long PCIe slot that creates the image shown on the monitor. Needed for gaming, video editing and 3D work.
6	**Power Supply Unit (PSU)**	A metal box (usually at the bottom or top) that converts wall AC power into the low DC voltages the parts need.
7	**HDD (Hard Disk Drive)**	A storage drive with spinning magnetic disks. Large and cheap, but slower. Keeps your files permanently.
8	**SSD (Solid State Drive)**	A storage drive with no moving parts. Much faster than an HDD and often holds the operating system.
9	**M.2 NVMe SSD**	A small stick-shaped SSD fixed directly onto the motherboard. It is the fastest type of storage.
10	**Case Fans**	Fans at the front, rear or top that push cool air in and hot air out to keep the whole cabinet cool.
11	**CMOS Battery**	A coin-sized battery (CR2032) on the motherboard. It keeps the clock and BIOS settings when the PC is off.

No.	Component	Short description
12	**Expansion Cards**	Optional cards in PCIe slots, such as a sound card, Wi-Fi card or LAN card, that add extra features.
13	**Optical Drive**	A CD/DVD drive found in older PCs. Many new cases do not have one.
14	**Cables and Connectors**	Power cables (24-pin, CPU 8-pin), SATA data cables, front-panel wires (power button, USB, audio) and fan cables that join everything together.

QUESTION 2 Disassembling a desktop PC: step by step

Follow these steps in order. Work slowly, keep screws in a small box, and photograph the inside before you remove anything so you can copy the layout later.

STEP 1 Shut down and unplug all power

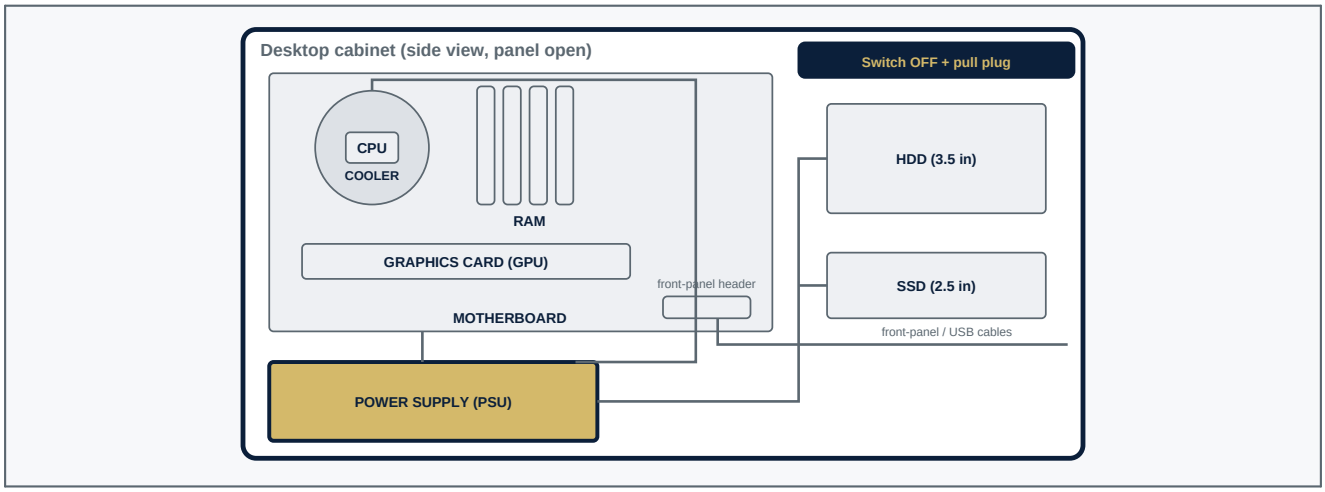

Fig. D1: Power supply switched off and the mains cable removed.

What I did: Closed all programs, shut the PC down from the operating system, switched the PSU switch at the back to **O (off)**, then pulled out the power cable. I also removed the monitor, keyboard, mouse and any other cables. Finally, I pressed the power button for about 5 seconds to drain leftover electricity.

> **Why it matters:** The PSU stores power even after shutdown. Working on a plugged-in PC can damage parts and give an electric shock.

STEP 2 Remove the side panel

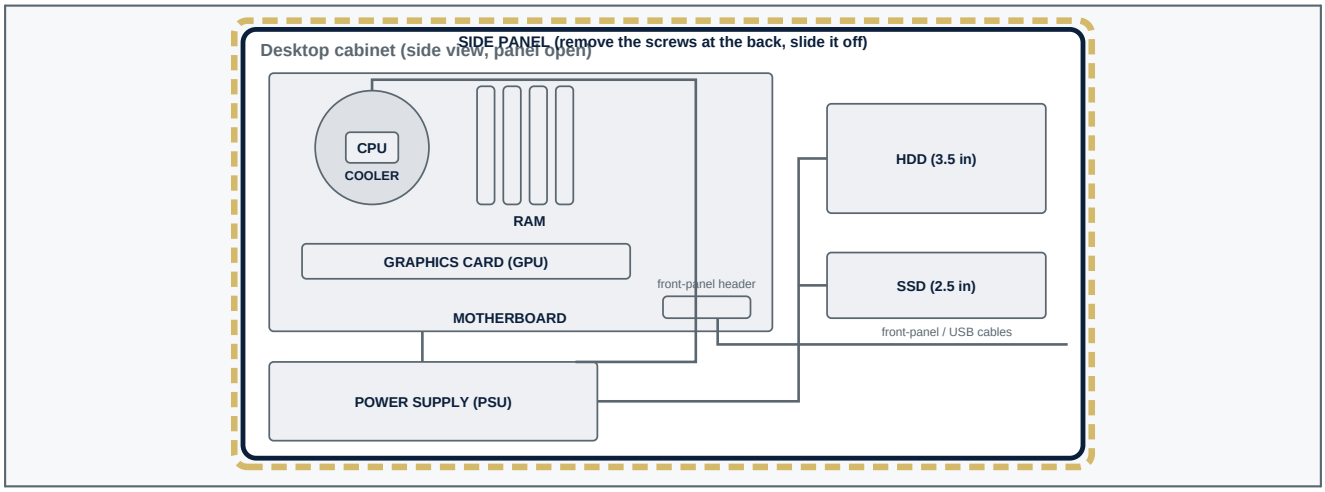

Fig. D2: The side panel (dashed gold outline) is held by screws at the back.

What I did: Placed the PC on a flat table, unscrewed the two thumbscrews (or Phillips screws) at the back edge, slid the side panel towards the rear and lifted it away. I kept the screws in a labelled container.

> **Tip:** Use the panel on the side that shows the motherboard. Many cases have a second panel on the other side for cable management.

STEP 3 Ground yourself and take reference photos

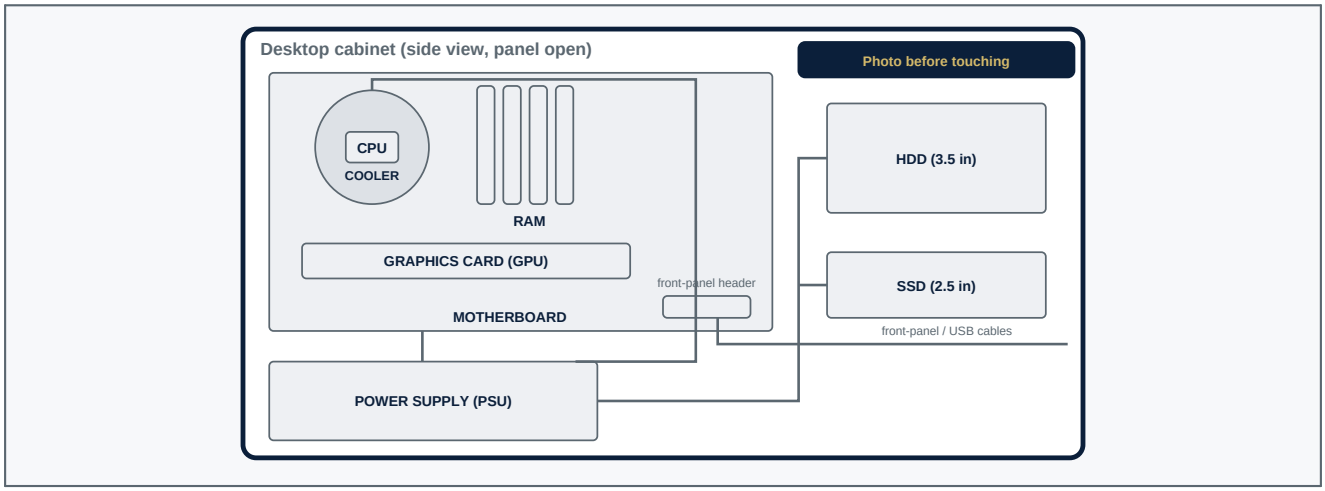

Fig. D3: Inside view photographed from the top and side before removing anything.

What I did: Put on an anti-static wrist strap clipped to the bare metal of the case (or touched the metal case frequently). I photographed the whole interior, especially the cables, and wrote small labels such as 'SATA 1' or 'Front USB'.

> **Why it matters:** Static electricity from your body can silently destroy chips. The photos work like a map when you reassemble.

STEP 4 Disconnect all cables

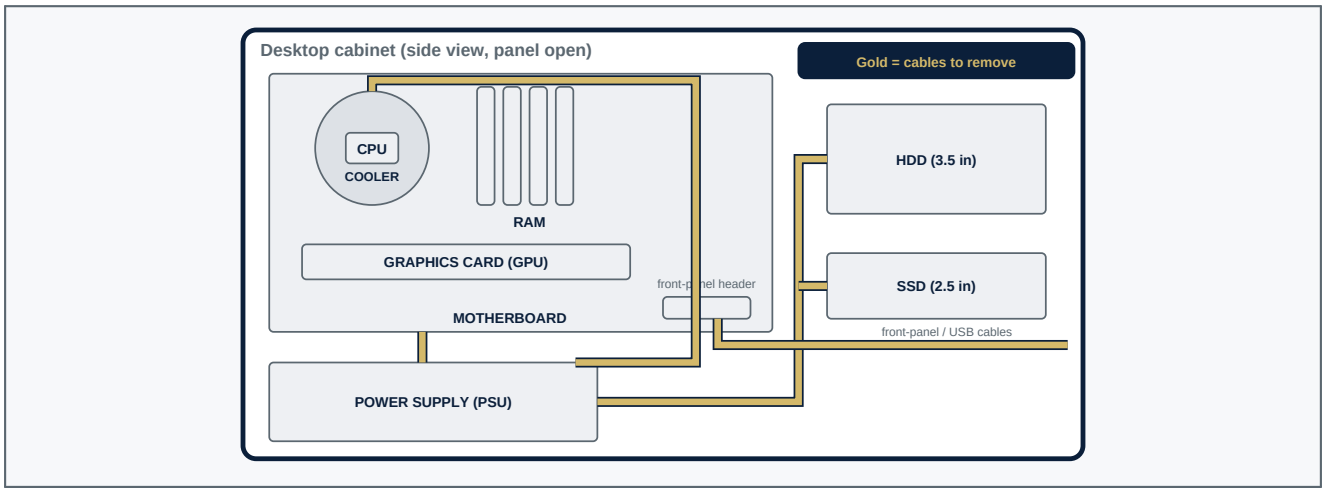

Fig. D4: Main power and data cables (gold lines) that must be unplugged.

What I did: Removed the 24-pin motherboard power, the 8-pin CPU power, the SATA power and data cables from the drives, GPU power cables (if any), fan cables, and the small front-panel connectors (power switch, reset, USB, audio). I pressed the clip on each plug and pulled the plug, not the wire.

> **Tip:** Pull on the plastic connector, never the wires. Do not force it: if a plug is stuck, look for a release clip.

STEP 5 Remove the graphics card (GPU)

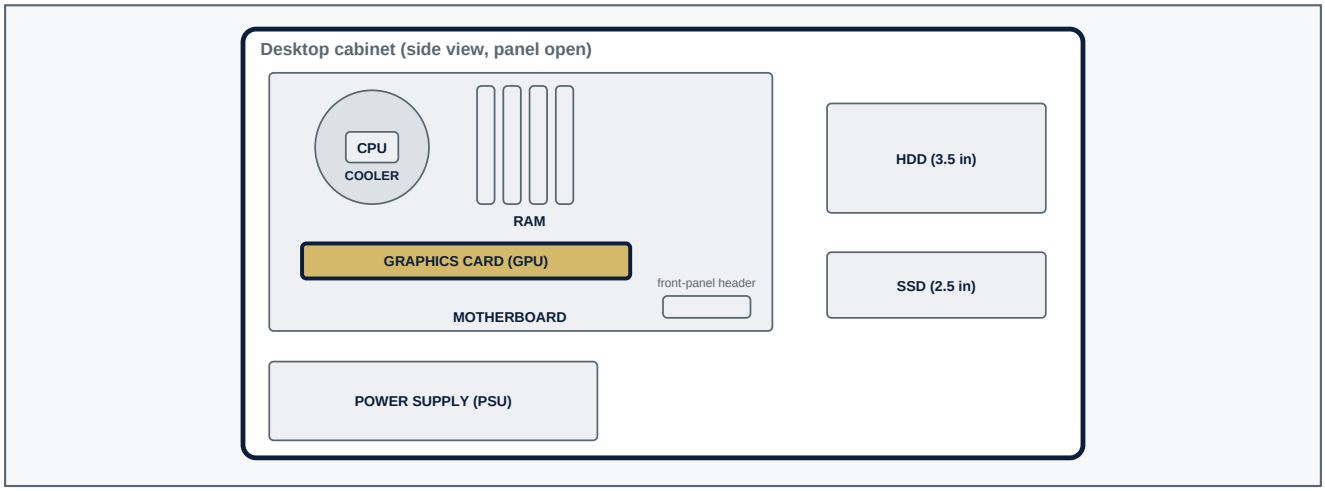

Fig. D5: Graphics card released from the long PCIe slot.

What I did: Unscrewed the card from the back bracket of the case, pushed the small release latch at the end of the PCIe slot, and gently pulled the card straight up and out. I placed it in an anti-static bag.

> **Tip:** Hold the card by its edges or metal bracket, not the gold contacts or chips.

STEP 6 Remove the RAM

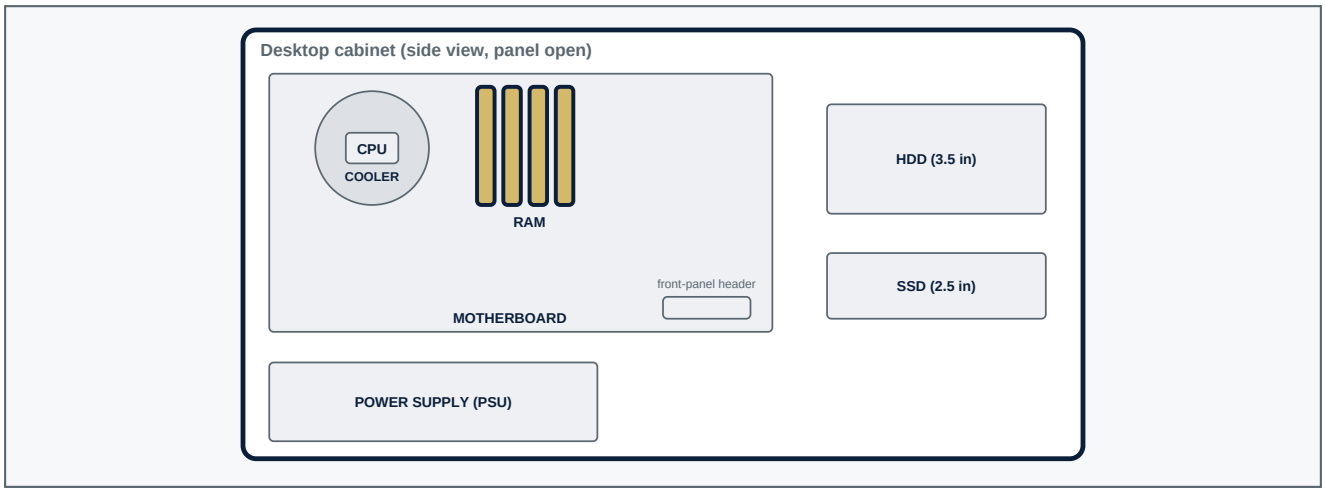

Fig. D6: RAM sticks released from their slots.

What I did: Pushed the plastic clips at both ends of each RAM slot outward. The stick pops up slightly, and I then lifted it out by its short edges and put it in an anti-static bag.

> **Tip:** Note which slots were used. Dual-channel PCs normally use slots 2 and 4, so remember the position.

STEP 7 Remove the HDD / SSD

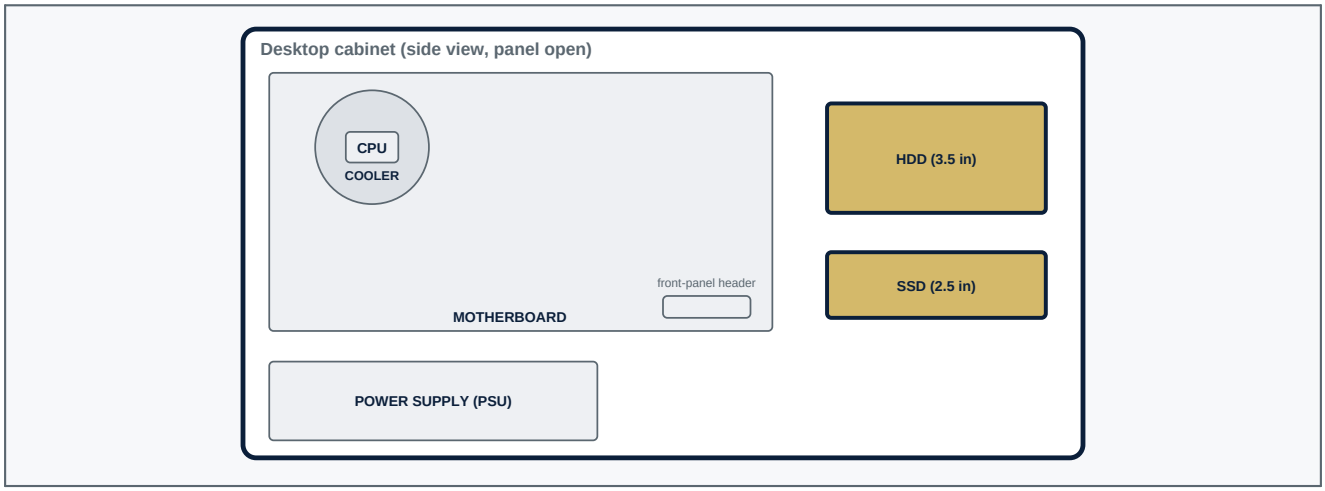

Fig. D7: Storage drives taken out of their bays.

What I did: Unscrewed the screws on the sides of the drive cage or sled (or released the tool-free clips) and slid the HDD (3.5 in) and SSD (2.5 in) out of the bays. For an M.2 SSD, I removed the single screw and pulled it out at an angle.

Why it matters: HDDs have moving parts, so avoid shaking or dropping them. Your data is stored inside, so handle them carefully.

STEP 8 Remove the CPU cooler and CPU (optional)

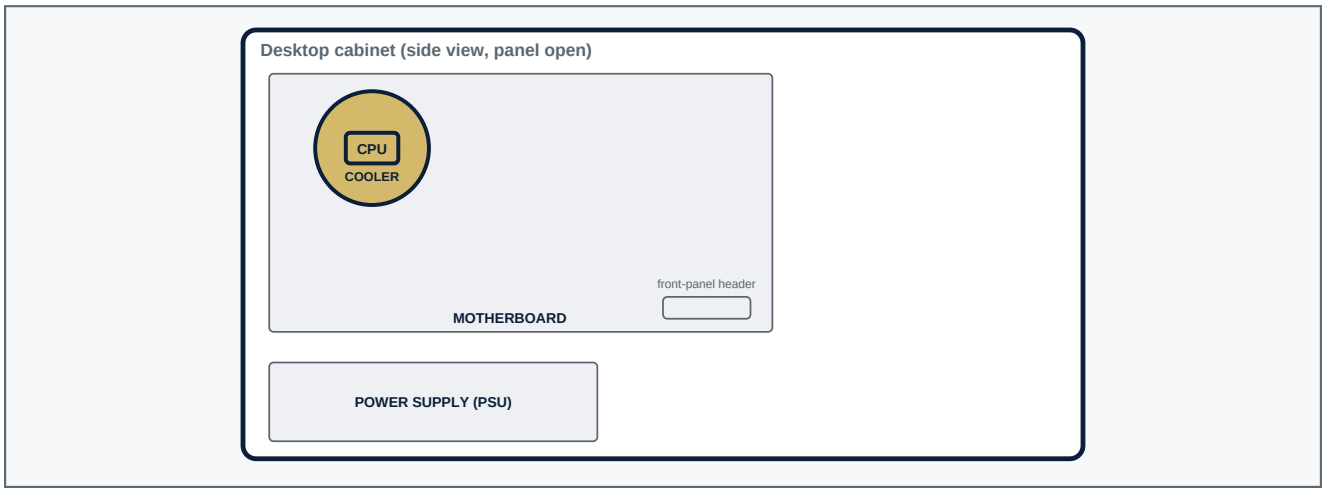

Fig. D8: Cooler released first, then CPU lifted from its socket.

What I did: Unplugged the cooler fan cable, loosened the cooler screws or clips in a cross pattern, twisted gently and lifted the cooler off. Then I lifted the CPU lever and removed the CPU by its edges, without touching the pins or gold pads.

Tip: Old thermal paste makes the cooler stick. Twist gently before pulling. Clean the paste off with isopropyl alcohol.

STEP 9 Remove the power supply (PSU)

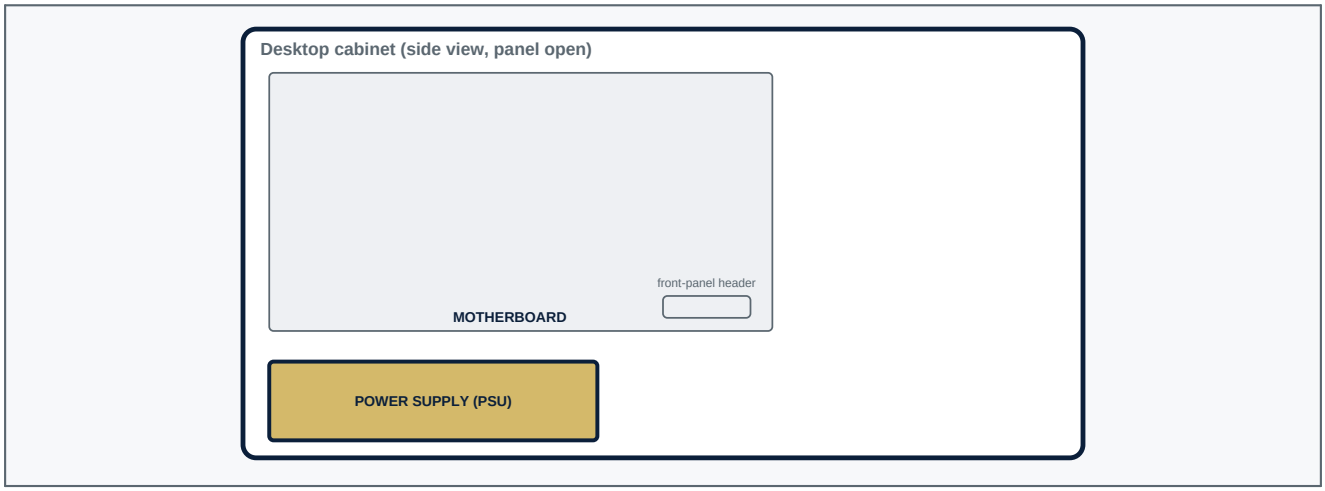

Fig. D9: PSU unscrewed from the back of the cabinet and slid out.

What I did: Removed the four screws at the back of the case that hold the PSU, supported it with one hand and slid it out of the case.

Tip: Never open the PSU box. It has high-voltage capacitors that can shock you even when unplugged.

STEP 10 Remove the motherboard

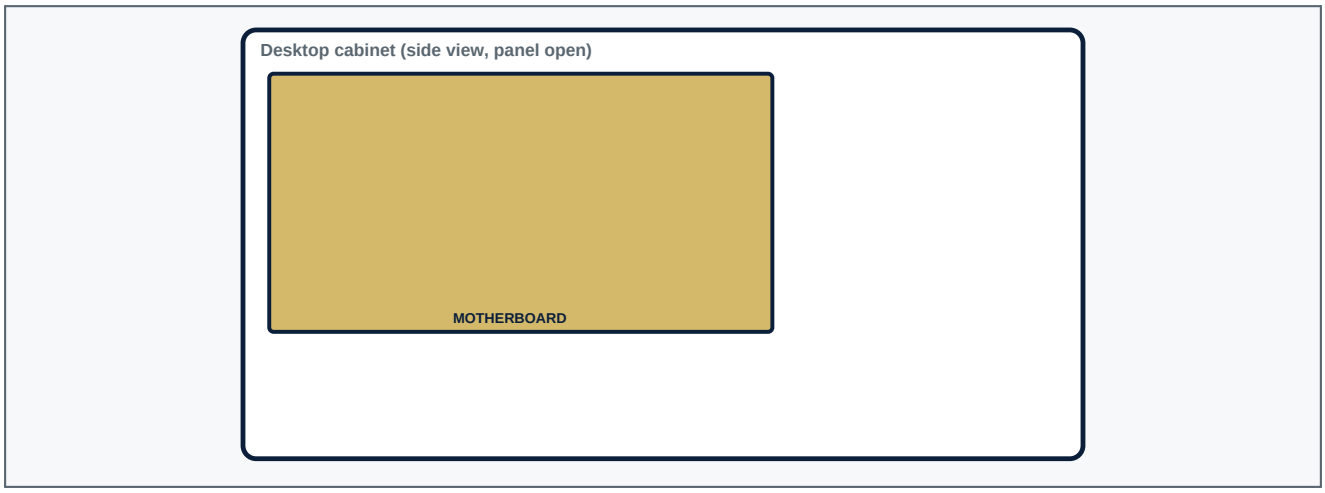

Fig. D10: Motherboard unscrewed from the standoffs and lifted out.

What I did: Took out all the screws (usually 6 to 9) that fix the board to the standoffs. Held it by the edges and lifted it out, taking care that the rear I/O ports cleared the metal back panel. I placed the board on an anti-static bag or its original box.

Why it matters: The board is thin and easy to crack or scratch. Never place it on bare metal or carpet.

QUESTION 3 Reassembling the PC: step by step

Reassembly is done in nearly the reverse order, with one important change: the CPU, cooler and RAM are fixed on the motherboard **before** it goes into the case, because there is much more space to work outside. In the diagrams, only the parts already installed are drawn, so you can see the PC being built up.

STEP 1 Prepare the empty case

Fig. R1: Empty case with standoff spacers screwed in where the motherboard will sit.

What I did: Laid the case flat, checked that it was clean and fixed the brass standoffs in the holes that match my motherboard size (ATX, micro-ATX, etc.). I also placed the back I/O shield if the board needed one.

> **Why the order matters:** The standoffs lift the board off the metal case. If they are missing or in the wrong place, the board can short-circuit and burn.

STEP 2 Install CPU and cooler on the motherboard

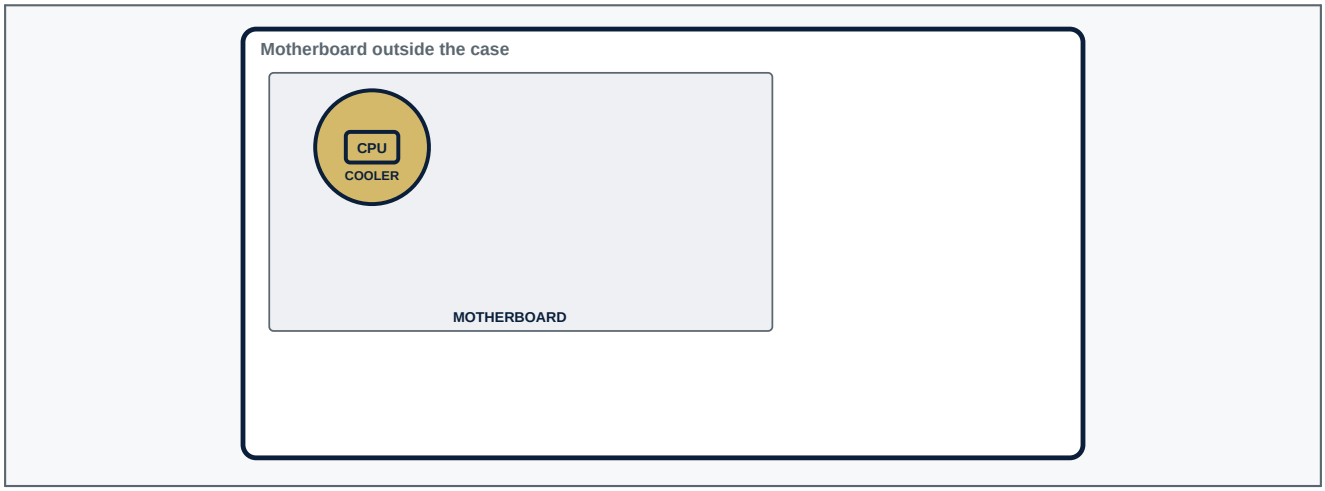

Fig. R2: CPU placed in its socket, cooler fixed on top with thermal paste.

What I did: Placed the motherboard on its anti-static bag. Lifted the CPU lever, matched the small triangle on the CPU with the one on the socket, lowered the CPU in without pressing, and closed the lever. Added a pea-sized dot of thermal paste, set the cooler on top, tightened the screws in a cross pattern and plugged in the CPU fan header.

Why the order matters: Once the motherboard is inside the case, the cooler is hard to reach and mount. Doing it outside is safer for the pins.

STEP 3 Install the RAM

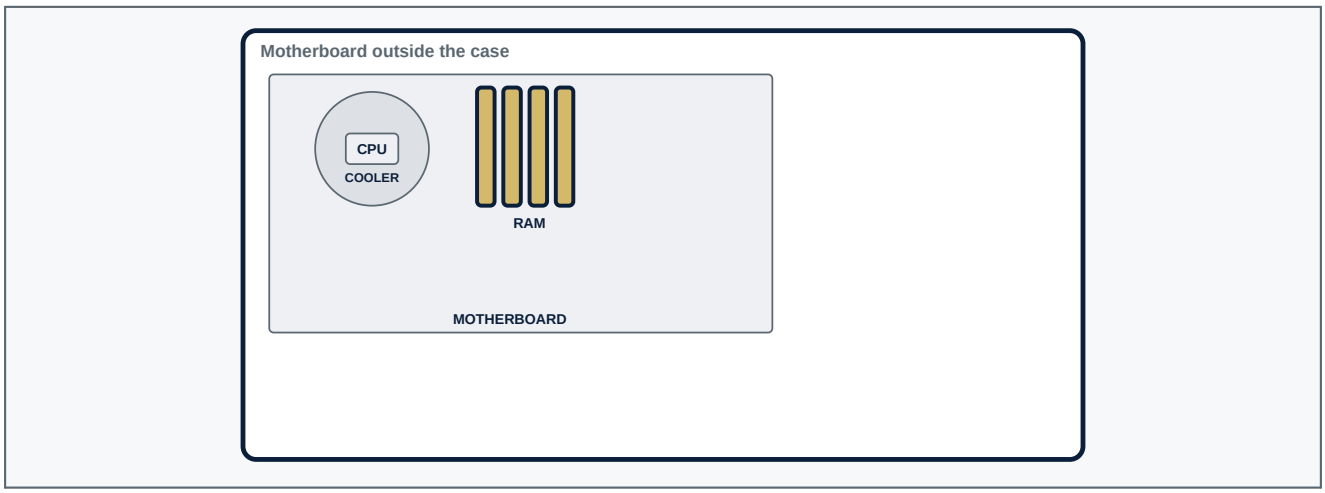

Fig. R3: RAM sticks pressed into the memory slots until the clips click.

What I did: Opened the clips of the RAM slots, matched the notch on the stick with the slot key and pressed both ends firmly until the clips clicked shut.

Why the order matters: RAM is easy to fit now, because the big GPU and the cables are not in the way yet. Wrong alignment can crack the slot, so use even pressure.

STEP 4 Mount the motherboard in the case

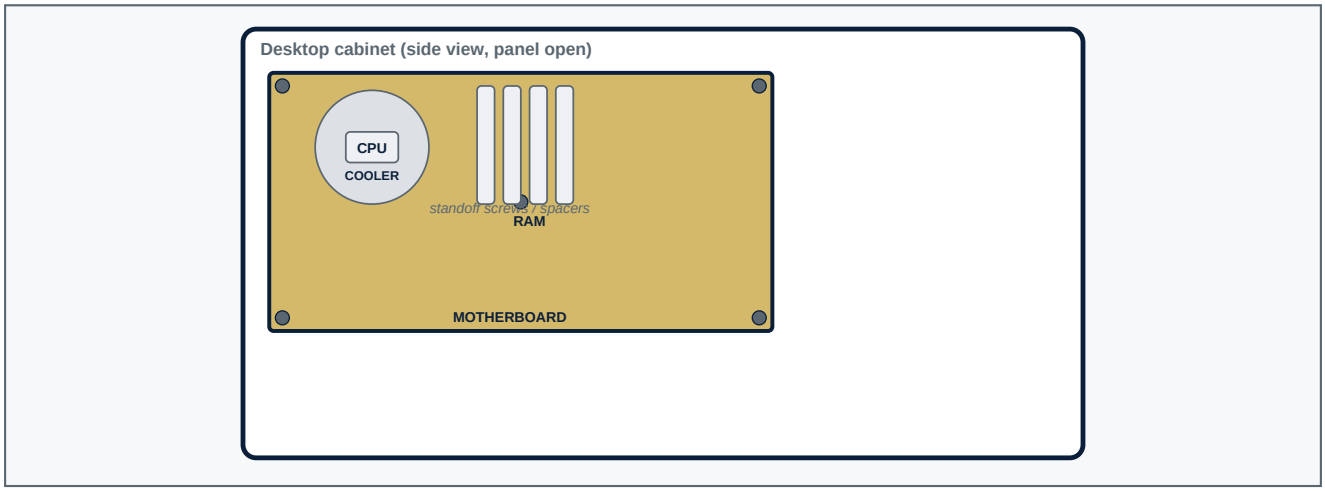

Fig. R4: Motherboard lowered onto the standoffs and screwed down.

What I did: Lowered the board so that the rear ports slid into the I/O shield and its holes lined up with the standoffs. Fixed it with screws, tightening each only until snug, not forcing them.

Why the order matters: The motherboard is the foundation. All other parts connect to it, so it must be fixed first.

STEP 5 Install the power supply (PSU)

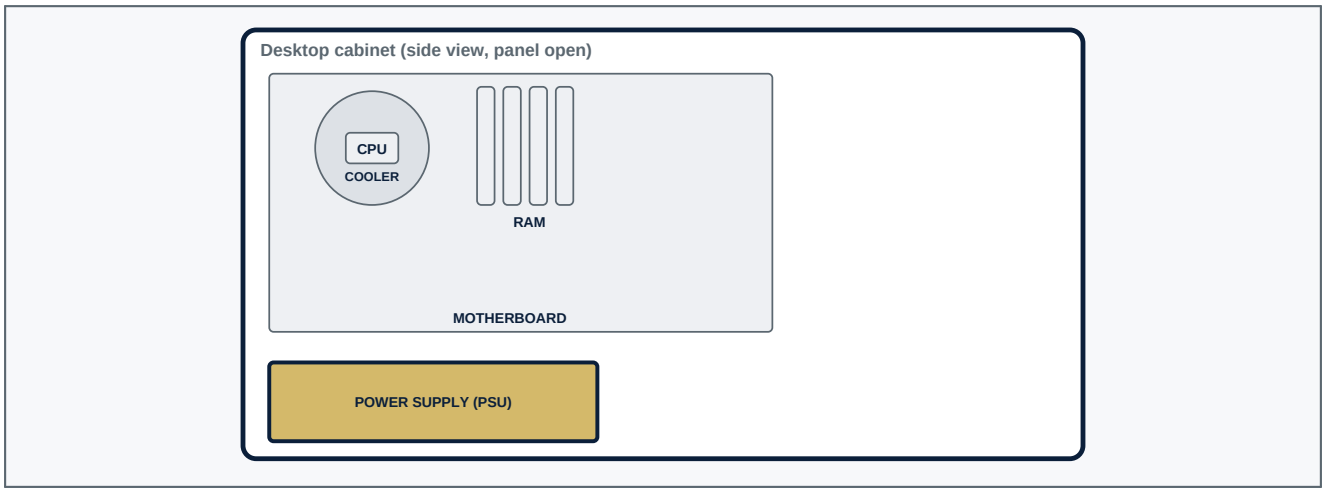

Fig. R5: PSU slid into its bay and fixed with four screws from the back.

What I did: Slid the PSU into its bay with the fan facing the vent, and fixed it with the four screws. I did not plug it into the wall yet.

Why the order matters: The PSU must be in position first so its cables can reach the motherboard, drives and GPU in the next steps.

STEP 6 Install the HDD / SSD

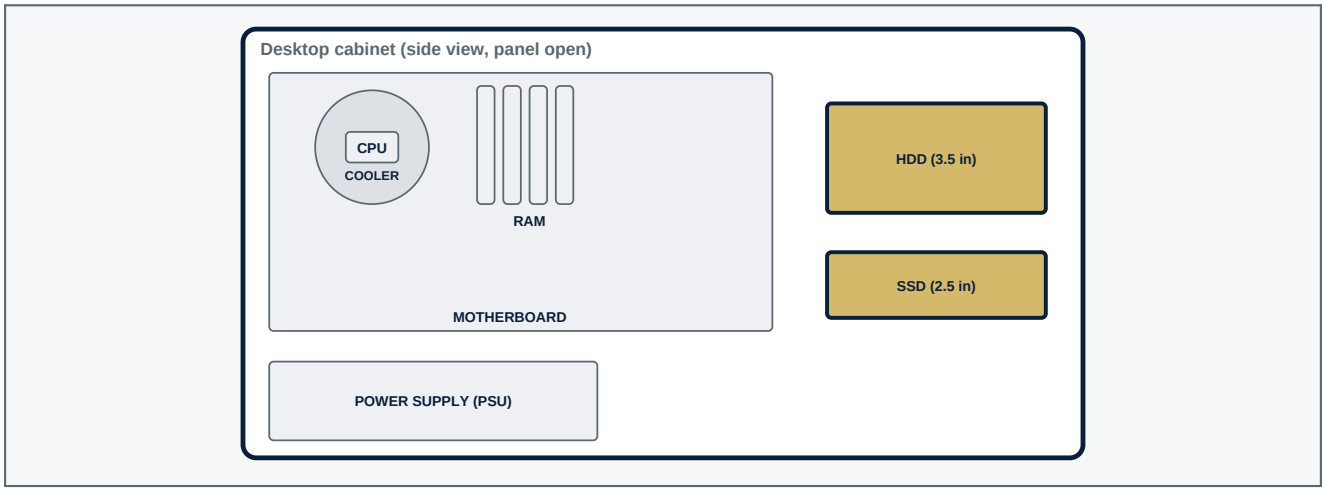

Fig. R6: Drives fixed in their bays.

What I did: Slid the 3.5 in HDD and 2.5 in SSD into their bays and fixed them with screws or clips, with the connector side facing the cables. A M.2 SSD goes into its slot and is held by a single screw.

Why the order matters: Drives are fitted before the GPU because the long graphics card can block access to the drive bays.

STEP 7 Install the graphics card (GPU)

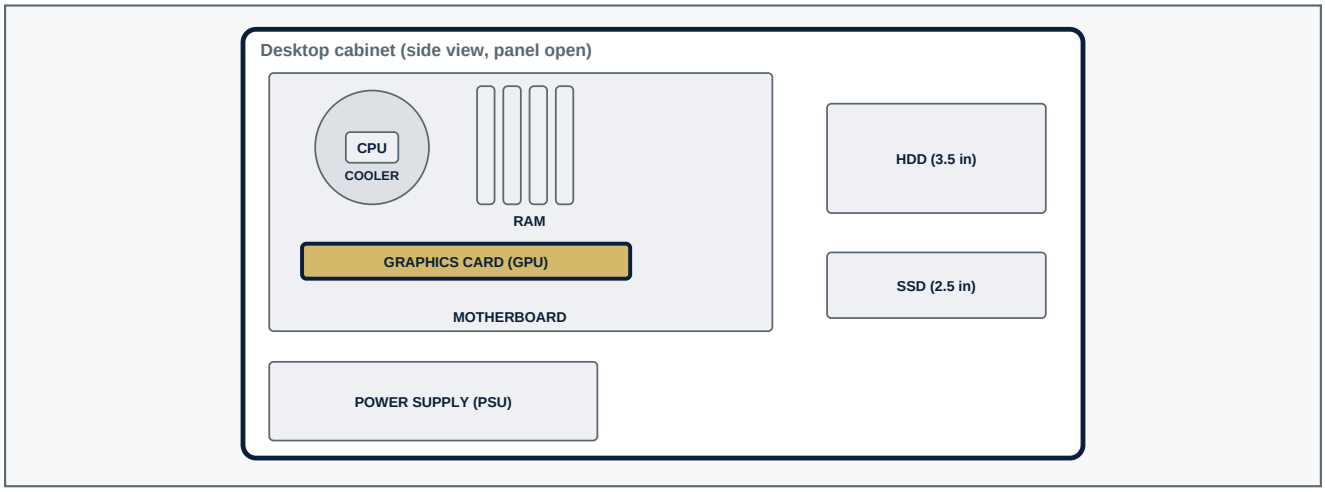

Fig. R7: GPU pressed into the PCIe slot and screwed to the case.

What I did: Removed the metal slot cover at the back, aligned the card with the top PCIe x16 slot and pushed down until the latch clicked. Fixed the bracket with a screw.

Why the order matters: The GPU is large and covers many connectors. If it is fitted early, it blocks access to the power headers and the SATA ports.

STEP 8 Connect all the cables

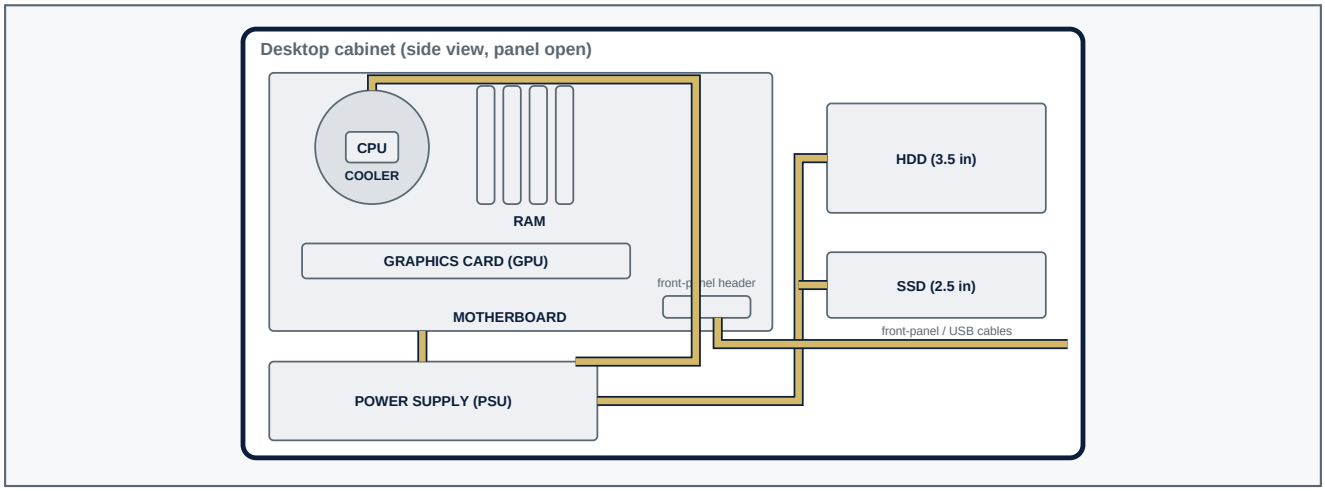

Fig. R8: All power and data cables reconnected (gold).

What I did: Connected: (1) the 24-pin main power, (2) the 8-pin CPU power, (3) the PCIe power to the GPU, (4) SATA power and data to the drives, (5) the front-panel connectors (power switch, reset, LEDs, USB, audio) following the motherboard manual, and (6) all fan cables. Then I tidied cables with zip ties or Velcro straps.

Why the order matters: Cables go in last, when everything is already mounted, so you can see where they must go. A loose or missing power cable is the most common reason a PC does not start.

STEP 9 Close the panel, reconnect and test

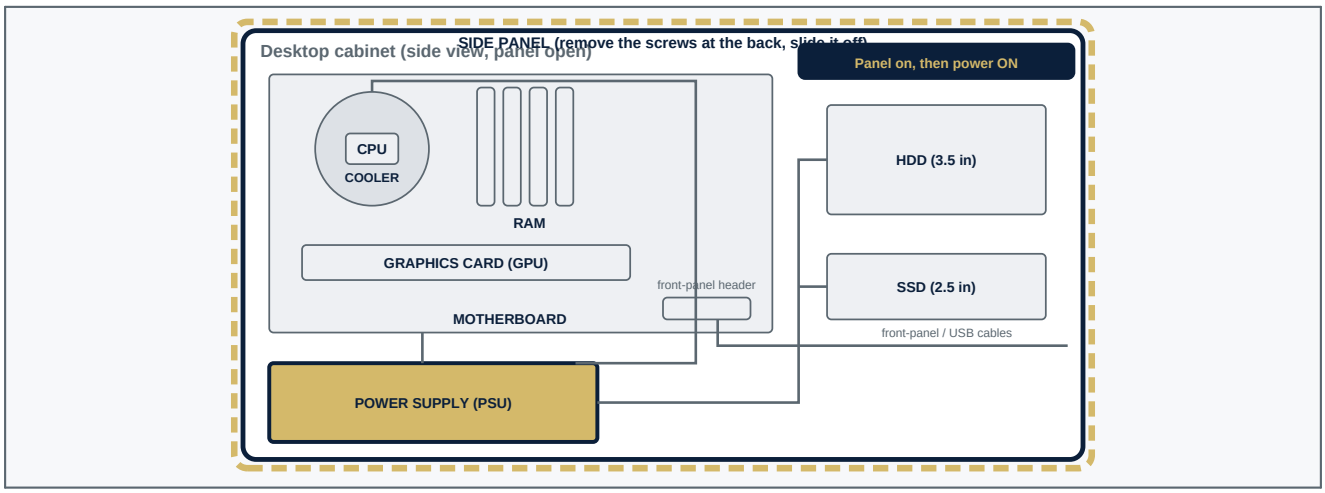

Fig. R9: Side panel refitted, monitor, keyboard, mouse and power cable connected.

What I did: Double-checked every connection and made sure no tool or screw was left inside. Refitted the side panel, connected the monitor, keyboard, mouse and power cable, switched the PSU to **I (on)** and pressed the power button. The PC showed the BIOS screen and booted into the operating system.

Why the order matters: The panel goes on only after a check, because it is much harder to fix a loose cable once the case is closed. Always test with the panel on only after you are sure everything is secure.

Remember: Summary of the order: Case standoffs > CPU & cooler > RAM > motherboard in case > PSU > storage > GPU > cables > panel & test. The golden rule: install the small, delicate parts when there is plenty of room, and the large parts (GPU) and cables only at the end.

QUESTION 4

Checklist: safety precautions and tools

Tick every box before you begin. Each item is short, but each has a real reason behind it.

A. Safety precautions

	Safety precaution	Why it is important
☐	**Shut down and unplug the PC**	Removes the risk of electric shock and prevents short circuits while parts are being touched.
☐	**Wear an anti-static wrist strap (or touch bare metal often)**	Static discharge you cannot even feel can permanently damage the CPU, RAM and motherboard.
☐	**Work on a clean, flat, non-metal table**	A stable table prevents falls. Bare metal or carpet can cause shorts or static.
☐	**Use bright light and clear space**	You can read tiny labels and see each connector, so mistakes are less likely.
☐	**Remove rings, watches and loose jewellery**	Metal can cause short circuits and scratch the parts, and it also slows your work.
☐	**Hold parts by the edges only**	Fingers carry oil and static. Touching chips, pins or gold contacts can damage them.
☐	**Never open the PSU or touch its insides**	PSU capacitors hold dangerous high voltage even after unplugging.
☐	**Keep drinks and food away**	A spill on live or even unplugged electronics can ruin parts.
☐	**Take photos and label cables first**	Gives you a reference to rebuild correctly, avoiding wrong connections.
☐	**Do not force any part or cable**	Connectors fit only one way. Forcing them bends pins and cracks boards.
☐	**Keep screws sorted in small containers**	Different screws are used for the case, drives and board. Mixing them can strip threads or lose them.
☐	**Keep children and pets away**	Small parts can be swallowed, and people moving around raise the risk of drops.

B. Tools and materials needed

	Tool / material	Why it is important
☐	**Phillips (cross) screwdriver, #1 and #2**	Almost every PC screw is a Phillips type. A correct size avoids stripped screw heads.
☐	**Small flat-head screwdriver**	Helps to release latches or to gently pry open stubborn clips.
☐	**Anti-static wrist strap and anti-static bags**	Protects parts from static and gives safe storage for removed components.

	Tool / material	Why it is important
☐	**Magnetic screw tray or small boxes**	Stops screws from rolling away and keeps them in order.
☐	**Zip ties or Velcro straps**	Keeps cables tidy so that air can flow and cables do not touch the fans.
☐	**Thermal paste**	A fresh thin layer between CPU and cooler moves the heat properly. It is needed if the cooler was removed.
☐	**Isopropyl alcohol (90%+) and lint-free cloth**	Cleans old thermal paste from the CPU and cooler without leaving residue.
☐	**Tweezers or needle-nose pliers**	Picks up dropped screws and handles tiny jumpers or connectors.
☐	**Torch or phone light**	Lights dark corners of the case so you can see connector labels.
☐	**Marker, masking tape and a phone camera**	To label cables and to record every step as evidence for your assignment.
☐	**Motherboard manual**	Shows exactly where the front-panel, fan and RAM connectors go on your board.
☐	**Soft brush or compressed air (optional)**	Removes dust while the case is open, which helps cooling and part life.

Conclusion: Disassembling and reassembling a PC is safe when you slow down, unplug everything, handle parts gently and follow the correct order. A few minutes spent on the checklist can save you from the cost of damaged hardware.